Supplementary Information

Towards Green Acetylene Synthesis and Application

Zihao Zhang, Abhinandan Nabera, Gonzalo Guillén-Gosálbez, Javier Pérez-Ramírez*

Institute for Chemical and Bioengineering, Department of Chemistry and Applied Biosciences, ETH Zurich, Vladimir-Prelog-Weg 1, 8093 Zurich, Switzerland.

* Corresponding author. E-mail: jpr@chem.ethz.ch

Table of Contents

Supplementary Notes 2

Supplementary Tables 4

Supplementary References 15

Supplementary Notes

Supplementary Note 1 | Life cycle assessment methodology

The global average climate change impacts are quantified using an attributional life cycle assessment (LCA) in accordance with ISO 14044/14040 standards.[1,2] We consider 1 kg of vinyl chloride monomer (VCM) or acetylene (C_2H_2) produced as the functional unit, following a cradle-to-gate approach that encompasses all activities from raw material acquisition to production. This approach assumes that the end-use phase is identical regardless of the synthesis route. The mass and energy flows for the VCM and C_2H_2 production routes were obtained from literature, as shown in **Supplementary Tables 3** and **6**, respectively. This data allows us to model the foreground system (VCM or C_2H_2 production plant), while data for the background system (activities supplying inputs to the production facility) were primarily sourced from the Ecoinvent v3.10 database.[3] We consider the 100-year average global warming potentials (GWPs) as implemented in the IPCC 2021 methodology.[4] The inventories used to calculate the climate change impact for all VCM and C_2H_2 production technologies are presented in **Supplementary Tables 4** and **7**, respectively. Notably, we assume that the biochar and calcium carbide production facilities are in close proximity to the acetylene production plant; therefore, we do not consider the negligible impact from transportation.

Furthermore, to calculate the climate change impacts for the year 2050, we conducted a prospective LCA. In this context, we considered the most optimistic future scenario aligned with the 1.5 °C climate target. To update the background system, we utilized future projections derived from Integrated Assessment Models (IAMs). IAMs, such as IMAGE (Integrated Model to Assess the Global Environment), provide a comprehensive understanding of the interactions among the biosphere, society, and the climate.[5] We used the *premise* v2.1.1.[6] framework to construct future background data consistent with the results from the IMAGE model. Additionally, the IMAGE model assumes specific Shared Socioeconomic Pathways (SSPs) that describe the relationship between challenges for adaptation and mitigation. In our study, we assumed the SSP2 scenario ("middle-of-the-road"). For the mitigation scenario, we considered the Representative Concentration Pathway (RCP) aimed at limiting the global mean surface temperature to 1.5 °C (RCP19 from *premise*). All calculations for the environmental assessment were conducted using the Brightway2 framework.[7]

We also perform an uncertainty assessment for the current scenarios using Monte Carlo sampling with 500 iterations, generated from the ecoinvent pedigree matrix that provides uncertainty levels for the life cycle inventory parameters. However, we do not perform an uncertainty assessment for the prospective 2050 scenarios due to the lack of available uncertainty data (*i.e.*, the corresponding pedigree matrix).

Supplementary Note 2 | Techno-economic assessment methodology

The mass and energy flows for the VCM and C_2H_2 production routes were obtained from literature, as shown in **Supplementary Tables 3** and **6**, respectively. The economic assessment considers only operational expenditures for the reference year 2022. Generally, capital costs are much lower than raw

material and utilities consumption expenses;[8] therefore, we neglect capital and transportation expenditures, providing a lower bound for the production costs in our analysis. **Supplementary Tables 5** and **8** present the prices of feedstock and utilities used in the production cost calculations for all the VCM and C_2H_2 production technologies, respectively. We also conduct an uncertainty assessment by considering both high and low costs for the feedstock and utilities needed in our process (as detailed in **Supplementary Tables 5** and **8**). Unlike the prospective LCA performed in this work, we do not conduct an economic assessment for 2050 due to the significant challenges posed by the high uncertainty and volatility of future energy prices, which are influenced by geopolitical scenarios and technological advancements.

Supplementary Note 3 | Safety risks and challenges of acetylene production technologies

Acetylene is a highly unstable compound under normal conditions. It reacts with oxygen at room temperature, producing water and carbon dioxide while releasing significant heat (1255.6 kJ mol^{-1}).[9] Its tendency to detonate makes it unsuitable for storage as a gas. Consequently, various safety risks and challenges in acetylene production must be addressed, depending on the synthesis technology. In the calcium carbide-based acetylene production route, calcium carbide is a highly flammable solid. Moreover, contact with moisture can form flammable acetylene gas and lime where the reaction heat can ignite acetylene.[10] Similarly, the partial combustion route involves significant safety risks and challenges, including high temperatures during preheating and preventing premature ignition due to the reactive oxygen-hydrocarbon mixture. Soot formation, which can block the reactor and destabilize the system, is also a major concern.[11] Thermal and non-thermal plasma processes for acetylene production pose distinct safety risks. Thermal plasma processes involve high temperatures, thereby increasing risks of explosions from flammable gases. On the other hand, non-thermal plasma operates at lower temperatures but also require high voltages, leading to electricity hazards and the risk of uncontrolled reactions. Therefore, it is essential to implement proper safety measures to maintain a safe and sustainable acetylene production process.

Supplementary Tables

Supplementary Table 1 | Comparison of representative results of thermal and non-thermal (cold and warm) plasma-driven methane coupling processes. Power supply, applied voltage, catalyst type, feed type, CH_4 conversion, selectivity to C_2 products, energy requirement (ER), specific energy input (SEI), and plasma temperature (thermal, cold, warm) are provided for comparison.

Plasma type	Power supply / W	Applied voltage / kV	Catalyst	Feed	Conv. / %	Sel. / %			ER / kWh kg^{-1} C_2H_2 (SEI / kJ mol^{-1} CH_4)	Plasma temp.	Ref
						C_2H_2	C_2H_4	C_2H_6			
Hüls arc	8000k	8	/	CH_4[a]	71	73	5	~	12.1 (275)	thermal	12,13
DuPont arc	9000k	~	/	CH_4/H_2	90	80	~	~	8.8	thermal	14
LC arc[b]	850	<12	/	CH_4/H_2	70	90	4	<1	10.2	thermal	15
corona	8.2	10.9	/	CH_4	28	76	~	~	26.2 (3403)	cold	16
corona	8.4	7	NaY	$CH_4/H_2/O_2$	38	32	2	7	~	cold	17
corona	8.4	7	/	$CH_4/H_2/O_2$	12	0	0	26	~	cold	17
pulse corona	30	20-32	Al_2O_3	CH_4/CO_2	43	23	5	3	~	cold	18
pulse corona	30	20-32	La_2O_3/Al_2O_3	CH_4/CO_2	25	55	9	9	~	cold	18
pulse corona	~	18-36	/	CH_4	45	68	~	~	~ (1788)	cold	19
DBD[c]	2	20	Fe-MFM-300	CH_4	3	38	58	0	~ (44.8)	cold	20
DBD	2	20	/	CH_4	1	17	47	22	~ (44.8)	cold	20
DBD	3.3	6	MgO/Al_2O_3	CH_4	23	28	30	27	~	cold	21
DBD	3.4	6	MgO/Al_2O_3	CH_4	16	22	26	36	~	cold	21
DBD	3.5	6	MgO/Al_2O_3	CH_4	10	11	19	48	~	cold	21
DBD	7-8	7.6	/	CH_4	32	7	6	20	~	cold	22
DBD	7-8	10	$BaTiO_3$	CH_4	6	18	32	20	~	cold	22
DBD	7-8	9	γ-Al_2O_3	CH_4	30	11	12	13	~	cold	22

DBD	7-8	8.9	α-Al_2O_3	CH_4	33	12	10	12	~	cold	22
DBD	7-8	9.8	SBA15	CH_4	20	10	18	15	~	cold	22
DBD	~	17	/	CH_4	12	3	3	31	~	cold	23
DBD	~	17	1g Pt/Al_2O_3	CH_4	12	2	2	49	~	cold	23
DBD	~	17	3g Pt/Al_2O_3	CH_4	19	~	<1	31	~	cold	23
DBD	36	<30	Pt/CeO_2	CH_4	36	6	7	47	~	cold	24
DBD	72	<30	Pt/CeO_2	CH_4	43	2	3	45	~	cold	24
DBD	>150	~	TiO_2/ZSM-5	CH_4/O_2	18	7	16	9	~	cold	25
DBD	>150	~	TiO_2/silicalite	CH_4/O_2	8	10	16	11	~	cold	25
pulsed	25	~	/	CH_4	44	95	3	<1	~	cold	26
pulsed	25	~	/	CH_4/O_2	59	59	~	~	~	cold	26
pulsed	25	~	/	$1CH_4/1H_2$	46	94	4	1	~	cold	26
pulsed	25	~	/	$1CH_4/9H_2$	12	64	15	18	~	cold	26
pulsed	~	21	/	CH_4	51	72	9[f]		~ (250)	cold	27
pulsed spark	~	~	/	CH_4	39	83	4	~	12.7	warm	28
pulsed spark	~	12.3	/	CH_4	24	85	5	2	9.2 (86)	warm	29
microwave	120	~	/	CH_4/H_2	51	61	12	~	73.8 (1076)	warm	30
microwave	130	~	Cu	CH_4	38	86	<1	2	~	warm	31
microwave	130	~	Cu/W	CH_4	22	82	1	3	~	warm	31
microwave	130	~	Cu/Ni	CH_4	17	76	2	4	~	warm	31
microwave	600	~[d]	/	CH_4	34	59	2	~	24.0 (224)	warm	32
microwave	600	~[d]	/	$3CH_4/1H_2$	43	62	2	~	25.0 (224)	warm	32
microwave	600	~[d]	/	$1CH_4/1H_2$	49	70	2	~	28.3 (224)	warm	32

microwave	600	~[e]	/	CH_4	31	46	8	~	33.2 (224)	warm	32
microwave	600	~[e]	/	$3CH_4/1H_2$	36	47	10	~	38.5 (224)	warm	32
microwave	600	~[e]	/	$1CH_4/1H_2$	42	60	10	~	38.3 (224)	warm	32
microwave	off	~	Mo/ZSM-5	CH_4	10	0	15	3	~	warm	33
microwave	110	~	Mo/ZSM-5	CH_4	11	35	10	2	~	warm	33
microwave	16	~	/	CH_4	24	41	~	~	143 (652)	warm	34
microwave	34	~	/	CH_4	57	57	~	~	44.7 (674)	warm	34
microwave	48	~	/	CH_4	80	61	~	~	28.1 (659)	warm	34
microwave	70	~	/	CH_4	94	71	~	~	20.0 (627)	warm	34
microwave	81	~	/	CH_4	93	71	~	~	20.8 (644)	warm	34

[a]Natural gas is used as the feed;

[b]Low-current arc;

[c]DBD: dielectric barrier discharge;

[d]110 mbar;

[e]55 mbar;

[f]Total selectivity to C_2H_4 and C_2H_6;

~ : not mentioned;

/ : not used.

Supplementary Table 2 | Comparison of representative results of thermal pyrolysis (TP) processes under various reaction conditions.

T / °C	GHSV / L g_{cat}^{-1} h^{-1} (h^{-1})	Catalyst	Feed	Conv. / %	Sel. / %			Total sel. to C_2 / %	Ref
					C_2H_2	C_2H_4	C_2H_6		
1100	24	Mo_2C	CH_4	23	5	13	3	21	35
1100	24	WC	CH_4	10	6	18	8	32	35
1000	12	Li/MgO	CH_4	5	1	3	6	10	36
865	~	FeS_2	CH_4/S_2	7-10	1-2	~	~	20-37	37
700	302	Mo/ZSM-5	CH_4	0.05	20	65	5	90	38
950	6	CaO-Na-Cl	CH_4/CO_2	19	9	63	22	94	39
950	6	CaO	CH_4/CO_2	9	5	64	31	100	39
200-1500	(10^5)	Pt/Al_2O_3	CH_4/O_2	63	7-15	4-5	<1	~	40
925	12	Ni–P/SiO_2	CH_4	1	6	64	18	88	40
900	~	Ni–P/SiO_2	C_2H_6	100	20	22	~	~	41
1020	(9400)	/	CH_4	5	18	28	6	52	42
1020	(9400)	CRS[a]	CH_4	8	19	26	4	49	42
1020	(9400)	Fe©CRS	CH_4	7	19	28	5	52	42
1020	(9400)	Fe/CRS	CH_4	8	19	25	3	47	42
1020	(9400)	Fe-CRS	CH_4	6	18	27	5	50	42
1020	(9400)	Fe/SiO_2	CH_4	7	18	27	5	50	42
1020	(9400)	Fe©CRS	CH_4/H_2	6	29	53	4	86	42
1020	1	/	CH_4	4	11	28	5	44	43
1020	1	SiO_2	CH_4	11	13	25	2	40	43
1020	1	Fe-SiO_2	CH_4	9	15	29	3	47	43
1020	1	CoFe-SiO_2	CH_4	10	13	26	2	41	43
1020	1	NiFe-SiO_2	CH_4	10	14	27	2	43	43
1020	1	PdFe-SiO_2	CH_4	8	14	29	3	46	43
1020	1	PtFe-SiO_2	CH_4	10	14	25	2	41	43
950	~	Fe reactor	CH_4/THN[b]	9	7	28	2	37	44
970	~	Fe reactor	CH_4/THN	11	8	27	2	37	44
990	~	Fe reactor	CH_4/THN	14	9	24	2	35	44
1010	~	Fe reactor	CH_4/THN	17	10	22	1	33	44

1030	~	Fe reactor	CH_4/THN	19	10	20	1	31	44
1050	~	Fe reactor	CH_4/THN	26	10	16	1	27	44
975	6	/	CH_4	1	0	0	4	4	45
975	6	CeO_2	CH_4	7	2	7	2	11	45
975	6	Pt_n/CeO_2	CH_4	10	6	8	2	16	45
975	6	Pt_1/CeO_2	CH_4	14	35	33	7	75	45
950	3.2	Fe/SiO_2[c]	CH_4	5	14	59	12	85	46
1000	3.2	Fe/SiO_2[c]	CH_4	13	16	44	4	64	46
1050	3.2	Fe/SiO_2[c]	CH_4	21	14	40	4	58	46
1070	3.2	Fe/SiO_2[c]	CH_4	26	14	37	3	54	46
1050	3.2	Fe/SiO_2	CH_4	14	18	49	5	72	46
1050	3.2	Fe/SiO_2	CH_4/H_2	4	25	62	6	93	46
800	6	NiP/SiO_2	CH_4	<1	0	19	38	57	47
900	6	NiP/SiO_2	CH_4	3	4	35	5	44	47
1000	~	Fe/SiO_2	CH_4	11	7	22	2	31	48

[a]CRS: cristobalite;

[b]THN: 1,2,3,4-tetrahydronaphthalen;

[c]Hydrogen-permeable tubular membrane reactor;

~ : not mentioned;

/ : not used.

Supplementary Table 3 | Inputs and outputs of the acetylene to VCM (ATV) and ethylene-based balanced process (EBP) routes per kg of VCM produced.

Stream	**ATV**[49]	**EBP**[50]
Feedstock		
Acetylene / kg	0.42	-
Ethylene / kg	-	0.45
HCl / kg	0.58	-
Chlorine / kg	-	0.57
Utilities		
Electricity / kWh	-	0.02
Cooling water / kg	75.99	196.85
Refrigeration to –25 °C / MJ	0.39	-
Refrigeration to –50 °C / MJ	-	2.15
Heating – natural gas / kg	$82 \cdot 10^{-5}$	0.05
Emissions to water		
Water / kg	-	$44 \cdot 10^{-4}$
Emissions to air		
Carbon dioxide / kg	-	0.02

Supplementary Table 4 | Inventories used in the life cycle assessment (LCA) to calculate the climate change impact for all VCM production technologies.

Flow	Activity	Source
Ethylene	market for ethylene	Ecoinvent v3.10
Bio ethylene	market for bio ethylene	Yang et al.[51]
Acetylene	acetylene production, coal, calcium carbide (**Supplementary Table 6**)	This work
Bio acetylene	acetylene production, biochar, calcium carbide (**Supplementary Table 6**)	This work
Chlorine	market for chlorine, gaseous	Ecoinvent v3.10
HCl	market for hydrochloric acid, without water, in 30% solution state	Ecoinvent v3.10
Heating – natural gas	heat production, natural gas, at boiler condensing modulating >100kW	Ecoinvent v3.10
Electricity	market group for electricity, high voltage	Ecoinvent v3.10
Cooling water	market for water, deionized	Ecoinvent v3.10
Refrigeration –25 °C	refrigeration –25 °C	Luyben et al.[52]
Refrigeration –50 °C	refrigeration –50 °C	Luyben et al.[52]

Supplementary Table 5 | Cost parameters used in calculating the unitary production cost for VCM.

Flow	Average	Low	High	Ref
Ethylene / USD kg^{-1}	1.770	1.293	1.895	50
Bio ethylene / USD kg^{-1}	4.386	-	-	51
Acetylene / USD kg^{-1}	1.179	0.862	1.865	This work
Bio acetylene / USD kg^{-1}	1.553	0.963	2.481	This work
Chlorine / USD kg^{-1}	0.389	-	-	50
HCl / USD kg^{-1}	0.153	-	-	53
Heating – natural gas / USD kg^{-1}	0.285	0.193	0.393	54
Cooling water / USD kg^{-1}	$48 \cdot 10^{-6}$	-	-	55
Grid electricity / USD kWh^{-1}	0.093	0.067	0.151	55
Refrigeration to –25 °C / USD MJ^{-1}	0.019	-	-	52
Refrigeration to –50 °C / USD MJ^{-1}	0.043	-	-	52

All costs are reported for the year 2022 as the reference year. The production costs for acetylene (only operational expenditures) are calculated in this work, referring to the material and energy flows and cost parameters provided in **Supplementary Table 6** and **Supplementary Table 7**, respectively.

Supplementary Table 6 | Inputs and outputs of the calcium carbide (CaC_2), partial combustion (PC), thermal plasma, and non-thermal plasma routes per kg of acetylene (C_2H_2) produced.

Stream	CaC_2[56]	PC[56]	Thermal plasma[56,57]	Non-thermal plasma[28]
Feedstock				
Limestone / kg	5.00	-	-	-
Coke or biochar / kg	2.00	-	-	-
Natural gas or biomethane / kg	-	4.46	1.55	1.48
N-methyl-2-pyrrolidone / kg	-	0.01	0.01	-
Sodium hypochlorite / kg	0.05	-	-	-
Sodium hydroxide / kg	$16 \cdot 10^{-4}$	-	-	-
Utilities				
Electricity / kWh	10.81	3.30	12.10[a]	12.65
Cooling water / kg	12.50	3.10	6.00	-
Heating – steam / kg	-	4.50	4.00	-
Emissions to air				
Particulate matter / kg	$51 \cdot 10^{-5}$	$25 \cdot 10^{-5}$	$23 \cdot 10^{-5}$	-
Nitrogen oxides / kg	$78 \cdot 10^{-5}$	$48 \cdot 10^{-5}$	$45 \cdot 10^{-5}$	-
Carbon dioxide / kg	3.60	0.80	-	-
Volatile organic compounds (VOC) / kg	0.32	-	-	-
Non-methane volatile organic compounds (NMVOC) / kg	-	$30 \cdot 10^{-6}$	$36 \cdot 10^{-6}$	-

[a]The electricity consumption was updated to 12.10 kWh kg^{-1} C_2H_2, as the original reference used for the thermal plasma inventory (from Zhang et al.[56]) reported a lower amount than the average range in the literature.[13,28,57–59]

Supplementary Table 7 | Inventories used in the life cycle assessment (LCA) to calculate the climate change impact for all acetylene production technologies. The life cycle inventories employed for biomethane were based on *premise* v2.1.1[6] sourced from Ecoinvent v3.10, which provides prospective inventories for a range of technologies based on the results derived from the IMAGE Integrated Assessment Model[5] under the SSP2 pathway.

Flow	Activity	Source
Limestone	market for limestone, unprocessed	Ecoinvent v3.10
Coke	market for coke	Ecoinvent v3.10
Biochar	market for biochar (using poplar as the biomass source)	Peters et al.[60]
Natural gas	market group for natural gas, high pressure	Ecoinvent v3.10
Biomethane	biomethane production, from biogas upgrading, using amine scrubbing	IMAGE SSP2-RCP6-2020 Ecoinvent v3.10
N-methyl-2-pyrrolidone	market for N-methyl-2-pyrrolidone	Ecoinvent v3.10
Sodium hypochlorite	market for sodium hypochlorite, without water, in 15% solution state	Ecoinvent v3.10
Sodium hydroxide	market for sodium hydroxide, without water, in 50% solution state	Ecoinvent v3.10
Electricity	market group for electricity, high voltage	Ecoinvent v3.10
Cooling water	market for water, deionised	Ecoinvent v3.10
Heating – steam	market for steam, in chemical industry	Ecoinvent v3.10

Supplementary Table 8 | Cost parameters used in calculating the unitary production cost for acetylene.

Flow	Average	Low	High	Ref
Coke / USD kg^{-1}	0.084	0.067	0.117	61
Biochar / USD kg^{-1}	0.271	0.117	0.425	62
Methane / USD kg^{-1}	0.285	0.193	0.393	54
Biomethane / USD kg^{-1}	1.488	1.058	1.901	63
Heating – steam / USD kg^{-1}	0.021	-	-	55
Cooling water / USD kg^{-1}	$48 \cdot 10^{-6}$	-	-	55
Grid electricity / USD kWh^{-1}	0.093	0.067	0.151	55

All costs are reported for the year 2022 as the reference year.

Supplementary References

1. International Standards Organization. *ISO 14040:2006 Environmental management—Life cycle assessment—Principles and framework* (2006).
2. International Standards Organization. *ISO 14044:2006 Environmental management—Life cycle assessment—Requirements and guidelines* (2006).
3. Wernet, G. et al. The ecoinvent database version 3 (part I): overview and methodology. *Int. J. Life Cycle Assess*. **21**, 1218–1230 (2016).
4. IPCC. *Climate Change 2021: The Physical Science Basis. Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change* (Cambridge University Press, 2021).
5. Stehfest, E., van Vuuren, D., Bouwman, L. & Kram, T. *Integrated assessment of global environmental change with IMAGE 3.0: Model description and policy applications* (Netherlands Environmental Assessment Agency (PBL), 2014).
6. Sacchi, R. et al. PRospective EnvironMental Impact asSEment (premise): A streamlined approach to producing databases for prospective life cycle assessment using integrated assessment models. *Renew. Sustain. Energy Rev.* **160**, 112311 (2022).
7. Mutel, C. Brightway: An open source framework for life cycle assessment. *J. Open Source Softw*. **2**, 236 (2017).
8. Galán-Martín, Á. et al. Sustainability footprints of a renewable carbon transition for the petrochemical sector within planetary boundaries. *One Earth* **4**, 565–583 (2021).
9. Pässler, P. *et al. Acetylene*. in *Ullmann's Encyclopedia of Industrial Chemistry* (Wiley, 2011).
10. New Jersey Department of Health, Hazardous Substance Fact Sheet: Acetylene (RTK Substance Number: 0312). Available at: https://nj.gov/health/eoh/rtkweb/documents/fs/0312.pdf.
11. Bartholomé, E. The BASF-process for production of acetylene by partial oxidation of gaseous hydrocarbons. *Chem. Eng. Sci.* **3**, 94–104 (1954).
12. Boulos, M. I., Fauchais, P. & Pfender, E. *High-power plasma torches and transferred arcs. in Handbook of Thermal Plasmas* 905–960 (Springer International Publishing, 2023).
13. Scapinello, M., Delikonstantis, E. & Stefanidis, G. D. The panorama of plasma-assisted non-oxidative methane reforming. *Chem. Eng. Process. Process Intensif.* **117**, 120–140 (2017).
14. Bilera, I. V. & Lebedev, Yu. A. Plasma-chemical production of acetylene from hydrocarbons: history and current status (a review). *Pet. Chem.* **62**, 329–351 (2022).
15. Khoe Dinh, D. et al. Efficient methane-to-acetylene conversion using low-current arcs. *RSC Adv.* **9**, 32403–32413 (2019).
16. Beloqui Redondo, A., Troussard, E. & van Bokhoven, J. A. Non-oxidative methane conversion assisted by corona discharge. *Fuel Process. Technol.* **104**, 265–270 (2012).
17. Liu, C., Mallinson, R. & Lobban, L. Nonoxidative methane conversion to acetylene over zeolite in a low temperature plasma. *J. Catal.* **179**, 326–334 (1998).

18. Zhang, X., Dai, B., Zhu, A., Gong, W. & Liu, C. The simultaneous activation of methane and carbon dioxide to C_2 hydrocarbons under pulse corona plasma over $La_2O_3/\gamma\text{-}Al_2O_3$ catalyst. *Catal. Today* **72**, 223–227 (2002).
19. Zhu, A., Gong, W., Zhang, X. & Zhang, B. Coupling of methane under pulse corona plasma (I) in the absence of oxygen. *Sci. China Ser. B-Chem.* **43**, 208–214 (2000).
20. Ma, Y. et al. Direct conversion of methane to ethylene and acetylene over an iron-based metal–organic framework. *J. Am. Chem. Soc.* **145**, 20792–20800 (2023).
21. Kasinathan, P. et al. Plasma-enhanced methane direct conversion over particle-size adjusted MO_x/Al_2O_3 (M = Ti and Mg) catalysts. *Plasma Chem. Plasma Process.* **34**, 1317–1330 (2014).
22. Taheraslani, M. & Gardeniers, H. Coupling of CH_4 to C_2 hydrocarbons in a packed bed DBD plasma reactor: the effect of dielectric constant and porosity of the packing. *Energies* **13**, 468 (2020).
23. Kim, S. S., Lee, H., Choi, J. W., Na, B. K. & Song, H. K. Methane conversion to higher hydrocarbons in a dielectric-barrier discharge reactor with $Pt/\gamma\text{-}Al_2O_3$ catalyst. *Catal. Commun.* **8**, 1438–1442 (2007).
24. Liu, L., Das, S., Zhang, Z. & Kawi, S. Nonoxidative coupling of methane over ceria-supported single-atom Pt catalysts in DBD plasma. *ACS Appl. Mater. Interfaces* **14**, 5363–5375 (2022).
25. Han, Q. et al. Conversion of methane to C_2 and C_3 hydrocarbons over TiO_2/ZSM-5 core–shell particles in an electric field. *RSC Adv.* **9**, 34793–34803 (2019).
26. Kado, S., Sekine, Y. & Fujimoto, K. Direct synthesis of acetylene from methane by direct current pulse discharge. *Chem. Commun.* **24**, 2485–2486 (1999).
27. Lotfalipour, R., Ghorbanzadeh, A. M. & Mahdian, A. Methane conversion by repetitive nanosecond pulsed plasma. *J. Phys. D: Appl. Phys.* **47**, 365201 (2014).
28. Yao, S., Nakayama, A. & Suzuki, E. Acetylene and hydrogen from pulsed plasma conversion of methane. *Catal. Today* **71**, 219–223 (2001).
29. Yao, S. L., Suzuki, E., Meng, N. & Nakayama, A. A high-efficiency reactor for the pulsed plasma conversion of methane. *Plasma Chem. Plasma Process.* **22**, 225–237 (2002).
30. Zhang, J., Yang, Y., Zhang, J., Liu, Q. & Tan, K. Non-oxidative coupling of methane to C_2 hydrocarbons under above-atmospheric pressure using pulsed microwave plasma. *Energy Fuels* **16**, 687–693 (2002).
31. Radoiu, M. T., Chen, Y. & Depew, M. C. Catalytic conversion of methane to acetylene induced by microwave irradiation. *Appl. Catal. B: Environ.* **43**, 187–193 (2003).
32. Wnukowski, M. et al. Influence of hydrogen addition on methane coupling in a moderate pressure microwave plasma. *Fuel* **288**, 119674 (2021).
33. Julian, I., Ramirez, H., Hueso, J. L., Mallada, R. & Santamaria, J. Non-oxidative methane conversion in microwave-assisted structured reactors. *Chem. Eng. J.* **377**, 119764 (2019).
34. Heintze, M. & Magureanu, M. Methane conversion into acetylene in a microwave plasma: optimization of the operating parameters. *J. Appl. Phys.* **92**, 2276–2283 (2002).
35. Zhang, S. B. X. Y. et al. Role and dynamics of transition metal carbides in methane coupling. *Chem. Sci.* **14**, 5899–5905 (2023).

36. Zhang, S. B. X. Y. et al. Lithium promotes acetylide formation on MgO during methane coupling under non-oxidative conditions. *Angew. Chem. Int. Ed.* **62**, e202307814 (2023).
37. Liu, S. et al. “Soft” oxidative coupling of methane to ethylene: Mechanistic insights from combined experiment and theory. *Proc. Natl Acad. Sci. USA* **118**, e2012666118 (2021).
38. Razdan, N. K., Kumar, A., Foley, B. L. & Bhan, A. Influence of ethylene and acetylene on the rate and reversibility of methane dehydroaromatization on Mo/H-ZSM-5 catalysts. *J. Catal.* **381**, 261–270 (2020).
39. Zhang, Y. et al. CO_2 oxidative coupling of methane using an earth-abundant CaO-based catalyst. *Sci. Rep.* **9**, 15454 (2019).
40. Hohn, K. L., Witt, P. M., Davis, M. B. & Schmidt, L. D. Methane coupling to acetylene over Pt-coated monoliths at millisecond contact times. *Catal. Lett.* **54**, 113–118 (1998).
41. Dipu, A. L. et al. Direct nonoxidative conversion of methane to higher hydrocarbons over silica-supported nickel phosphide catalyst. *ACS Catal.* **10**, 375–379 (2020).
42. Han, S. J., Lee, S. W., Kim, H. W., Kim, S. K. & Kim, Y. T. Nonoxidative direct conversion of methane on silica-based iron catalysts: effect of catalytic surface. *ACS Catal.* **9**, 7984–7997 (2019).
43. Han, S. J. et al. Methane direct conversion to olefins, aromatics, and hydrogen over silica entrapped bimetallic MeFe-SiO_2 (Me = Co, Ni, Pd, Pt) catalysts. *Mol. Catal.* **535**, 112864 (2023).
44. Hao, J. et al. Enhanced methane conversion to olefins and aromatics by H-donor molecules under nonoxidative condition. *ACS Catal.* **9**, 9045–9050 (2019).
45. Xie, P. et al. Nanoceria-supported single-atom platinum catalysts for direct methane conversion. *ACS Catal.* **8**, 4044–4048 (2018).
46. Sakbodin, M., Wu, Y., Oh, S. C., Wachsman, E. D. & Liu, D. Hydrogen-permeable tubular membrane reactor: promoting conversion and product selectivity for non-oxidative activation of methane over an Fe©SiO_2 catalyst. *Angew. Chem. Int. Ed.* **55**, 16149–16152 (2016).
47. Dipu, A. L., Nishikawa, Y., Inami, Y., Iguchi, S. & Yamanaka, I. Development of highly active silica-supported nickel phosphide catalysts for direct dehydrogenative conversion of methane to higher hydrocarbons. *Catal. Lett.* **152**, 199–212 (2022).
48. Oh, S. C. et al. Direct non-oxidative methane conversion in a millisecond catalytic wall reactor. *Angew. Chem. Int. Ed.* **58**, 7083–7086 (2019).
49. Dattani, J., Devani, D. & Sahu, O. Cleaner production of vinyl chloride monomer (VCM). *Int. J. Sci. Eng. Res.* **4**, 701–705 (2013).
50. Medrano-García, J. D. et al. Economic and environmental competitiveness of ethane-based technologies for vinyl chloride synthesis. *ACS Sustain. Chem. Eng.* **11**, 13062–13069 (2023).
51. Yang, M., Tian, X. & You, F. Manufacturing ethylene from wet shale gas and biomass: comparative technoeconomic analysis and environmental life cycle assessment. *Ind. Eng. Chem. Res.* **57**, 5980–5998 (2018).

52. Luyben, W. L. Estimating refrigeration costs at cryogenic temperatures. *Comput. Chem. Eng.* **103**, 144–150 (2017).
53. Wang, Y. et al. A decarbonized vinyl chloride synthesis via selective oxidative coupling of methyl chloride. Preprint at https://doi.org/10.26434/chemrxiv-2024-vlq7m-v2 (2024).
54. Commodity markets. *World Bank*. Available at: https://www.worldbank.org/en/research/commodity-markets (2024).
55. Ioannou, I., D'Angelo, S. C., Martín, A. J., Pérez-Ramírez, J. & Guillén-Gosálbez, G. Hybridization of fossil- and CO_2-based routes for ethylene production using renewable energy. *ChemSusChem* **13**, 6370–6380 (2020).
56. Zhang, S. et al. Life cycle assessment of acetylene production from calcium carbide and methane in China. *J. Clean. Prod.* **322**, 129055 (2021).
57. Jiang, C. & Hu, J. *Conversion of Methane to Acetylene*. 93–116 (Royal Society of Chemistry, 2023).
58. Anderson, R. P., Fincke, J. R. & Taylor, C. E. Conversion of natural gas to liquids via acetylene as an intermediate. *Fuel* **81**, 909–925 (2002).
59. Shlyapin, D. A., Afonasenko, T. N., Glyzdova, D. V., Leont'eva, N. N. & Lavrenov, A. V. Acetylene production technologies in the 21st century: main trends of their development in the paradigm of low-carbon economy of the future. *Catal. Ind.* **14**, 251–267 (2022).
60. Peters, J. F., Iribarren, D. & Dufour, J. Biomass pyrolysis for biochar or energy applications? A life cycle assessment. *Environ. Sci. Technol.* **49**, 5195–5202 (2015).
61. Coal PRICE Today | Coal Spot Price Chart | Live Price of Coal per Ounce. *Markets Insider.* Available at: https://markets.businessinsider.com/commodities/coal-price (2024).
62. Peters, J. *Pyrolysis for biofuels or biochar? A thermodynamic, environmental and economic assessment*. PhD thesis (2015).
63. Market state and trends in renewable and low-carbon gases in Europe. *European Biogas Association*. Available at: https://www.europeanbiogas.eu/market-state-and-trends-in-renewable-and-low-carbon-gases-in-europe/.